

A Certified Admission Gate for Online Reconfiguration of Time-Sensitive Networks

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Preparatory work for the PhD thesis *Certified Dynamic Reconfiguration of Time-Sensitive Networks:
Online Admission Control under Worst-Case Determinism Guarantees*

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Outline

1. Problem
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1. Problem: Worst-Case Guarantees in a Changing Network

Context

Factories, vehicles and aircraft replace dedicated fieldbuses with **one Ethernet network** shared by control loops, sensors, video and best-effort traffic. For control traffic, a late message is a **failure**, not a slower service: what is needed is a **guaranteed worst case**.

- **TSN** (IEEE 802.1Q [1]) gives bounded latency, but only for a flow set **analysed offline**, with network calculus [2, 3] or a synthesised schedule [4].
- In operation, **flows are added while the network runs**; each new flow can delay flows already admitted, **even on other paths** that share a port.
- Fast **proposers** exist (heuristics, ILP/SMT, deep RL), but their outputs are **optimised, not proven**.

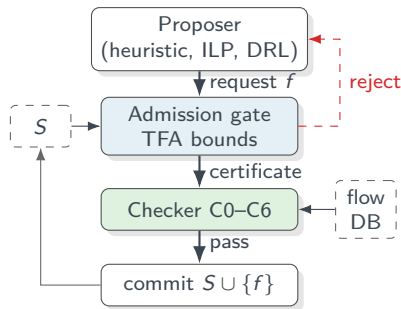
After this change, does every admitted flow still meet its deadline?

1. Problem: Our Approach and Contributions

A certified admission gate

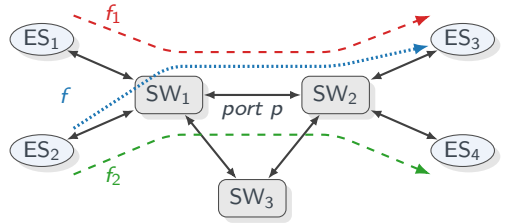
Placed between **any** proposer and the network, like a safety shield [5]: a change is committed only with a **certificate** that an **independent checker** re-verifies.

- **C1.** Gate $\text{admit}(S, f) \rightarrow (\text{verdict}, \text{certificate})$ for non-preemptive static-priority TSN, with a soundness argument.
- **C2.** Incremental gate: recomputes only the affected ports, **bit-identical** bounds.
- **C3.** Validation against **panco** [6], simulation and tests; fully reproducible.
- **C4.** Explicit limits of the guarantee.



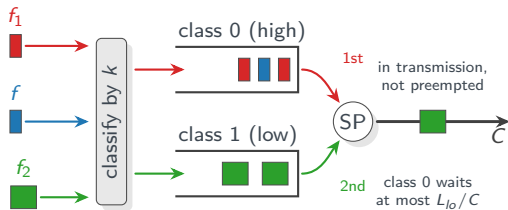
2. Model: Network, Flows and Scheduling

- **Network:** switches and end stations, full-duplex links of rate $C = 1 \text{ Gbit/s}$; each link direction is an output **port**; processing delay t_{proc} .
- **Routing:** fixed shortest paths; port dependency graph checked **feed-forward**.
- **Flow** f : token bucket $\alpha_f(t) = b_f + r_f t$, max frame L_f , priority class k_f (0 highest), deadline $D_f^{\max}$.
- **Ports:** non-preemptive static priority (**NP-SP**), FIFO within a class.



Admitted f_1 (class 0), f_2 (class 1) and a new class-0 request f share port p : admitting f can delay both.

2. Model: Packet Handling and Delay Bounds at an NP-SP Port



Per-hop bound for class k at port p

Residual rate-latency service [3, 7]:

$$d_k = \frac{B_H + L_{lo} + B_k}{C - R_H} + t_{proc}, \quad R_H + R_k \leq C$$

Burst propagation and end-to-end bound (TFA)

$$b' = b + r_f d_k, \quad D_f = \sum_{h \in \mathcal{P}_f} d_{k_f}^{(h)} \leq D_f^{\max}$$

B_H, R_H : higher-priority bursts and rates; L_{lo} : largest lower-priority frame (non-preemptive blocking); B_k, R_k : class- k aggregates.

3. Admission Gate: Certificate and Independent Checker

Interface and invariant

On request f , analyse $S \cup \{f\}$; accept iff it is schedulable. **Invariant:** every flow in S meets its deadline. A rejection lists **every** violated flow, not only f .

Certificate

Per flow: path and, per hop, b_{in} , $(B_H, R_H, L_{lo}, B_k, R_k)$, d ; end-to-end bound; fingerprint of S before/after (XOR of SHA-256 flow hashes).

Checker (about fifty lines, no analysis code)

- C0 one witness per admitted flow, with its own parameters
- C1 each path is the fixed route
- C2 first input burst $\geq b_f$
- C3 next input burst $\geq b_{in} + r_f d$
- C4 claimed aggregates $\geq$ those recomputed from all witnesses at the port
- C5 $R_H < C$, $R_H + R_k \leq C$, $d \geq$ per-hop bound
- C6 end-to-end bound $\geq \sum d$ and $\leq$ deadline

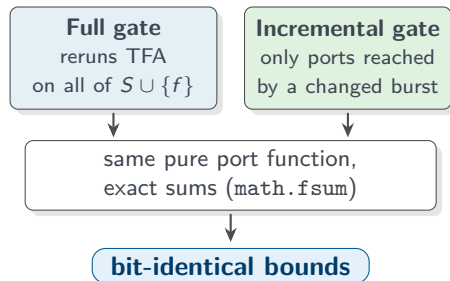
3. Admission Gate: Soundness and Incremental Computation

Proposition (soundness)

If a certificate passes C0–C6, every admitted flow meets its deadline under the model.

Why it holds

- The bound grows with B_H, R_H, L_{lo}, B_k .
- Induction over the **topological order of ports**: every witnessed burst over-approximates the true one (C2–C5).
- Summing the hops gives the deadline (C6).
- **C0 is necessary**: omitting a flow only *lowers* the others' aggregates; only the flow database reveals it.



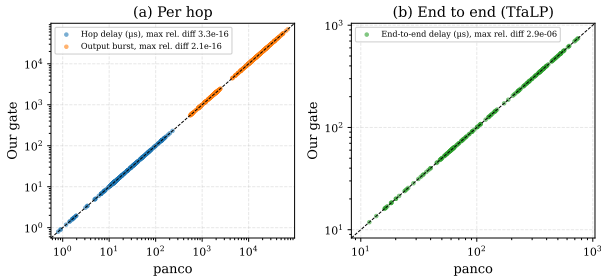
The incremental gate issues a **delta certificate** (changed flows only).

4. Results: Setup and Validation Against panco

Setup

Line	5 switches, 4 end stations each
Tree	1 core + 4 edge, 5 end stations each
Ports	48 per topology, $C = 1$ Gbit/s
Class 0	64–256 B, deadline 0.1–0.5 ms
Class 1	500–1500 B, deadline 0.5–2 ms
Runs	1,000 requests $\times$ 10 seeds, both gates

Reference: panco [6], per-hop residual servers and end-to-end TFA as a linear program.



$$3.3 \times 10^{-16}$$

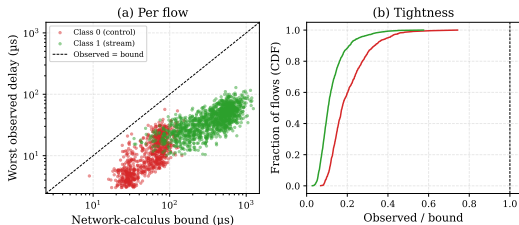
per hop, 837 values

$$2.9 \times 10^{-6}$$

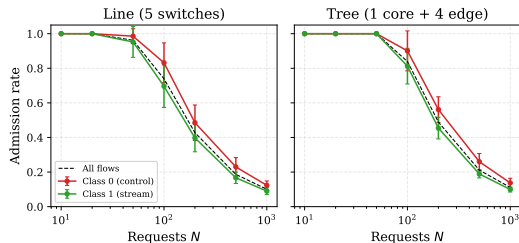
end to end, 210 values

max. relative difference (end to end: lp_solve prints 6 digits)

4. Results: Simulation and Admission Rate

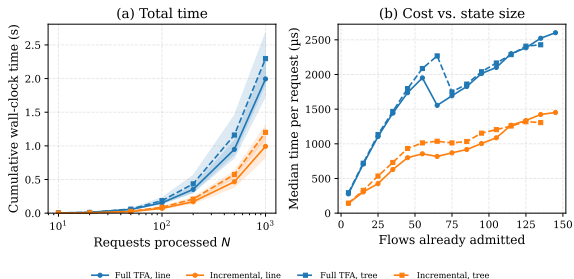


0 violations over 2,118 admitted flows; median observed/bound 0.17 (class 0), 0.11 (class 1). A sanity check, not a tightness measure.



All requests accepted up to ≈ 50 ; saturation at 100 ± 20 (line) and 111 ± 16 (tree) of 1,000; class 0 accepted slightly more often.

4. Results: Run Time of the Full and Incremental Gates



Topology	Gate	Median μs	Speed-up
Line	Full	1944	2.02×
	Incremental	964	
Tree	Full	2202	1.91×
	Incremental	1186	

- Identical verdicts and **bit-identical** bounds.
- Gain modest on small networks: a request still re-evaluates 19 of 48 ports (median).
- 18 unit tests, incl. tampered certificates (C5 and C0 reject them).

5. Limitations

Model

Single link rate, fixed routing, feed-forward, strict NP-SP; no CBS/TAS shapers, preemption or clock effects; Ethernet overheads folded into L_f, b_f, r_f .

Certification

Proof sketch on paper, not mechanised; floating-point checker with 10^{-9} tolerance; the certificate attests to the model, not to the deployed switches.

Analysis

TFA is sound but pessimistic compared with PMOO or LP-based analyses [8, 7]: fewer admissions, never an unsafe one.

Evaluation

Additions only; small networks (saturation ≈ 100 flows), two classes; simulator does not force the worst case; no proposer connected yet.

6. Conclusion

Contribution

- **Safe traffic allocation:** a flow is admitted only if every deadline still holds.
- **Certificate** per decision, re-checked by a 50-line checker (C0–C6).
- **Any proposer**, even a learned one, cannot commit an unsafe allocation.
- **Incremental gate:** ≈ 1 ms, bit-identical bounds, $1.9\text{--}2.0\times$ faster.
- **Validated:** matches panco; 0 violations over 2,118 flows.

Future directions

- **Capacity:** tighter, still checkable analyses than TFA.
- **Coverage:** CBS and TAS shapers, frame preemption.
- **Trust:** mechanised proof, exact-arithmetic checker.
- **Operations:** flow removal and batched changes.
- **Proposers:** the gate as a shield behind heuristic, ILP and DRL.

Thank you for your attention!

Questions?

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